The documentation and process conversion measures necessary to comply with this revision shall be completed by 23 July 1997.

INCH-POUND

MIL-PRF-19500/118E
23 April 1997
SUPERSEDING
MIL-S-19500/118D
11 August 1986

PERFORMANCE SPECIFICATION SHEET

SEMICONDUCTOR DEVICE, DIODE, SILICON,
TYPES 1N483B, 1N485B, 1N486B, 1N5194, 1N5194UR, 1N5194US, 1N5195, 1N5195UR, 1N5195US,
1N5196, 1N5196UR, AND 1N5196US JAN, JANTX, JANTXV, AND JANS

This specification is approved for use by all Departments
and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the performance requirements for silicon diodes. Four levels of product assurance are provided for each device type as specified in MIL-PRF-19500 (see 6.3.2).

1.2 Physical dimensions. See figure 1, figure 2, and figure 3 (see 3.3).

1.3 Maximum ratings.

Type 1/	V_{RM}	V_{RWM}	I_O	I_O 2/ T_A = 150°C	I_{FSM} t_p = 1/120 s T_A = 25°C	T_{op} and T_{STG}	I_F
	V (pk)	V (pk)	mA	mA	A	°C	mA dc
1N483B, 1N5194	80	70	200	50	2	-65 to +175	650
1N485B, 1N5195	180	180	200	50	2	-65 to +175	650
1N486B, 1N5196	250	225	200	50	2	-65 to +175	650

1/ The electrical characteristics for UR and US suffix devices are identical to their corresponding non-UR or US suffix devices unless otherwise noted (Applies only to the 1N5194, 1N5195, and 1N5196).
2/ Derate 1.2 mA/°C between 25°C and 150°C. Derate 1 mA/°C between 150°C and 175°C.

1.4 Primary electrical characteristics at T_A = +25EC, unless otherwise indicated.

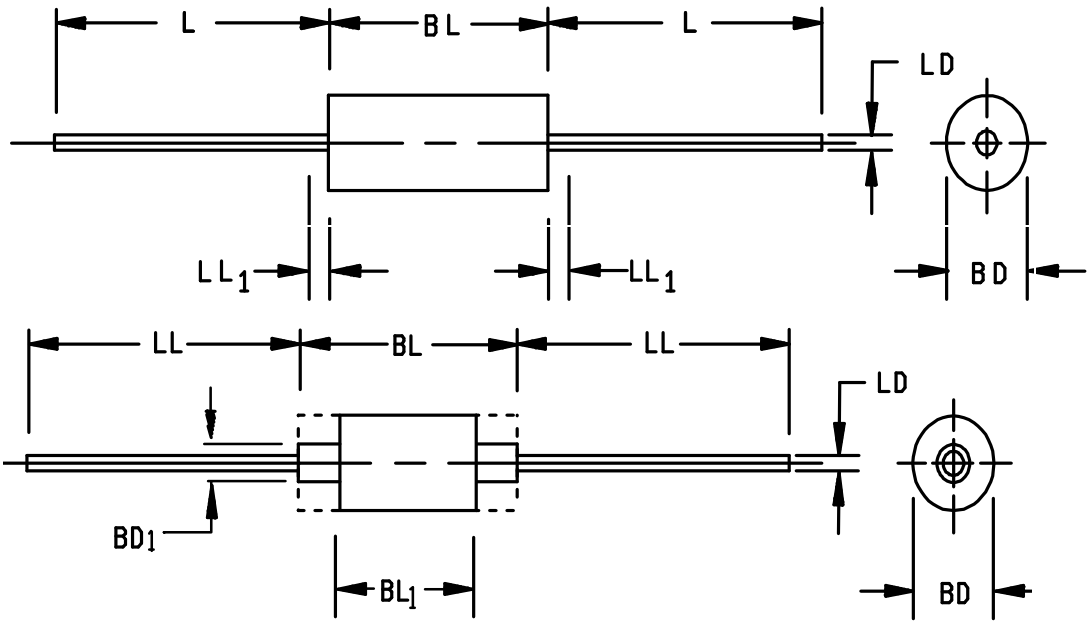
Type 1/	V_{F1}	I_{R1} at V_{RWM}	I_{R2} at V_{RWM} T_A = 150°C
	V dc	nA dc	:A dc
1N483B, 1N5194	1.0	25	5
1N485B, 1N5195	1.0	25	5
1N486B, 1N5196	1.0	25	5

1/ The electrical characteristics for UR and US suffix devices are identical to their corresponding non-UR or US suffix devices unless otherwise noted (Applies only to the 1N5194, 1N5195, and 1N5196).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Defense Supply Center Columbus, ATTN: DSCC-VAT, 3990 East Broad St., Columbus, OH 43216-5000, by using the addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5961
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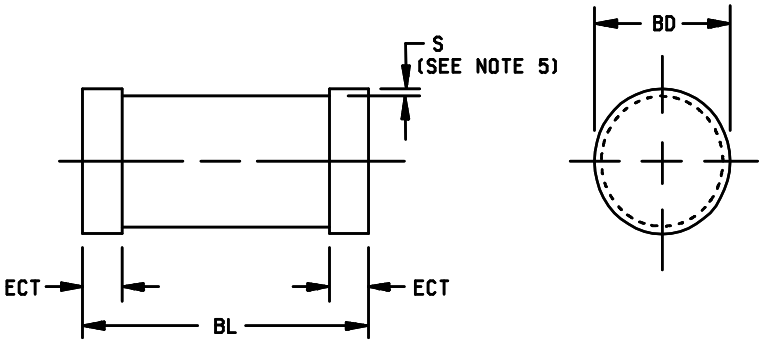


Type	Symbol	Dimensions				Notes
		Inches		Millimeters		
		Min	Max	Min	Max	
	LD	0.085	0.125	2.16	3.18	3
1N483B 1N485B 1N486B	BD	0.018	0.022	0.46	0.56	2,5
	BL	0.230	0.300	5.84	7.62	
	LL	1.00	1.50	25.40	38.10	4,5
	BD	0.056	0.094	1.42	2.39	
	BD1	0.030		0.76		
1N5194 1N5195 1N5196	BL	0.120	0.180	3.05	4.57	
	BL1	0.075		1.91		
	LD	0.019	0.021	0.48	0.53	2, 5
	LL	1.000		25.40		4, 5

NOTES:
1. Metric equivalents are given for general information only.
2. The specified lead diameter applies in the zone between .050 and 1.00 inch from the diode body. Outside this zone the lead diameter is not controlled.
3. The minimum body diameter shall be maintained over .15 inch of body length.
4. The leads shall be electrically insulated from the case.
5. Both leads.

FIGURE 1. Physical dimensions.

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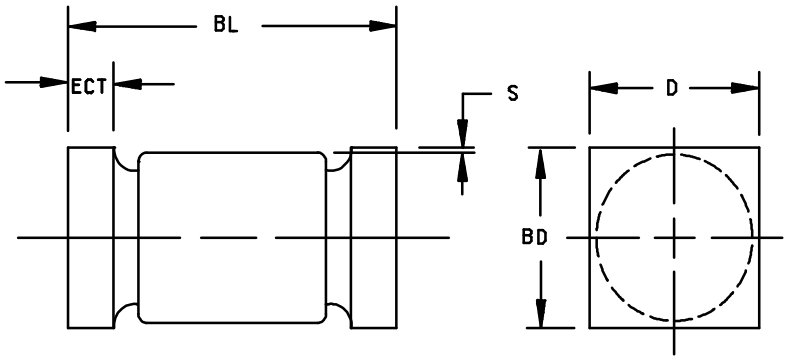


Symbol	Dimensions			
	Inches		Millimeters	
	Min	Max	Min	Max
BD	.063	.067	1.60	1.70
ECT	.016	.022	0.41	0.55
BL	.130	.146	3.30	3.70
S	.001 min		0.03 min	

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Dimensions are pre-solder dip.
4. Cathode marking to be either a color band or four dots spaced 90° apart.
5. Minimum clearance of glass body to mounting surface on all orientations.
6. In accordance with ANSI Y14.5M, diameters are equivalent to ϕx symbology.

FIGURE 2. Physical dimensions 1N5194UR, 1N5195UR, AND 1N5196UR.

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Symbol	Dimensions			
	Inches		Millimeters	
	Min	Max	Min	Max
BD	.091	.103	2.31	2.62
ECT	.019	.028	0.48	0.71
BL	.168	.200	4.27	5.08
S	.003 min		0.08 min	

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Dimensions are pre-solder dip.
4. Minimum clearance of glass body to mounting surface on all orientations.
5. Cathode marking to be either a color band or four dots spaced 90° apart.
6. In accordance with ANSI Y14.5M, diameters are equivalent to ϕx symbology.

FIGURE 3. Physical dimensions 1N5194US, 1N5195US, and 1N5196US.

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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

DEPARTMENT OF DEFENSE

MIL-PRF-19500 - Semiconductor Devices, General Specification for.

STANDARD

MILITARY

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Unless otherwise indicated, copies of the above specifications, standards, and handbooks are available from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated specifications or specification sheets), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Associated specification. The individual item requirements shall be in accordance with MIL-PRF-19500, and as specified herein.

3.2 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500.

3.3 Interface requirements and physical dimensions. The Interface requirements and physical dimensions shall be as specified in MIL-PRF-19500, and figures 1, 2, and 3 herein.

3.3.1 Lead finish. Lead finish shall be solderable as defined in MIL-PRF-19500, MIL-STD-750, and herein.

3.3.2 Diode construction. All devices shall be metallurgically bonded double plug construction in accordance with the requirements of category I, II, or III (see MIL-PRF-19500, paragraph 30.14 and subparagraphs).

3.3.3 JANS construction. JANS devices shall be metallurgically bonded, thermally matched, non-cavity, double plug construction per MIL-PRF-19500, Appendix A, paragraph 30.14.8. The use of silver button dumet package design or opaque glass is prohibited.

3.4 Marking. Marking shall be in accordance with MIL-PRF-19500.

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3.4.1 Polarity. The polarity shall be indicated with a contrasting color band to denote the cathode end. No color coding will be permitted.

3.4.2 Marking of UR and US version devices. For UR and US version devices only, all marking (except polarity) may be omitted from the body, but shall be retained on the initial container.

3.5 Electrical performance characteristics. Unless otherwise specified herein, the electrical performance characteristics are as specified in paragraph 1.3, 1.4, and table I.

3.6 Electrical test requirements. The electrical test requirements shall be the subgroups specified in paragraphs 4.4.2 and 4.4.3.

3.7 Qualification. Devices furnished under this specification shall be products that are authorized by the qualifying activity for listing on the applicable qualified products list before contract award (see 4.2 and 6.2).

4. VERIFICATION

4.1 Classification of Inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3)

c. Conformance inspection (see 4.4).

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500.

4.3 Screening (JANTX, JANTXV, and JANS levels only). Screening shall be in accordance with MIL-PRF-19500 (table IV), and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen (see table IV of MIL-PRF-19500)	Measurement	
	JANS	JANTX and JANTXV levels
3c	Thermal impedance (see 4.3.2)	Thermal impedance (see 4.3.2)
9	I_{R1}, V_{F1}	Not applicable
11	I_{R1}, V_{F1} ; ΔI_{R1} = 100% of initial reading or 10 nA dc, whichever is greater. ΔV_F = ± 50 mV dc change from initial reading	I_{R1}, V_{F1}
12	See 4.3.1	See 4.3.1
13 1/	Subgroup 2 of table I herein; ΔI_{R1} = 100% of initial reading or 10 nA dc, whichever is greater ΔV_{F1} = 50 mV dc change from initial reading	Subgroup 2 of table I herein; ΔI_{R1} = 100% of initial reading or 10 nA dc, whichever is greater ΔV_{F1} = 50 mV dc change from initial reading

1/ Thermal impedance need not be repeated at this step.

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4.3.1 Burn-in test. All devices shall be operated under one of the following conditions:

Type 1/, 2/	T_A = +25° C, ±3° C I_O = 200 mA f = 60 Hz	T_A = +25° C, ±3° C
1N483B, 1N5194, 1N5194UR, 1N5194US	V_{RWM} = 70 V(pk)	I_F = 175 mA
1N485B, 1N5195, 1N5195UR, 1N5195US	V_{RWM} = 180 V(pk)	I_F = 175 mA
1N486B, 1N5196, 1N5196UR, 1N5196US	V_{RWM} = 225 V(pk)	I_F = 175 mA

1/ Derate at 1.2 mA/° C between +25° C and +150° C. Derate at 1.0 mA/° C above +150° C.

2/ Electrical characteristics for UR and US suffix devices are identical to the corresponding non-UR or US suffix devices unless otherwise specified. (Applies only to the 1N5194, 1N5195, and 1N5196).

4.3.2 Thermal impedance $Z_{\Theta JX}$ measurements for screening. Thermal impedance $Z_{\Theta JX}$ measurements shall be performed in accordance with MIL-STD-750, method 3101. The maximum limit (not to exceed the group A, subgroup 2 limit) for $Z_{\Theta JX}$ in screening (table IV of MIL-PRF-19500) shall be derived by each vendor by means of statistical process control. When the process has exhibited control and capability, the capability data shall be used to establish the fixed screening limit. In addition to screening, once a fixed limit has been established, monitor all future sealing lots using a random five piece sample from each lot to be plotted on the applicable X, R chart.

4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500, and as specified herein.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-PRF-19500, and table I herein. Electrical measurements (end points) shall be in accordance with the table I, group A, subgroup 2 herein. The following test conditions shall be used for $Z_{\Theta JX}$, group A inspection:

a. I_M measurement current- - - - - - - - - - - 1 mA - 10 mA
b. I_H forward heating current- - - - - - - - - .5 A - 1.0 A
c. t_H heating time - - - - - - - - - - - - - - 10 ms
d. t_{MD} measurement delay time - - - - - - - - 100 us (max)

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table VIa (JANS) and table VIb (JAN, JANTX and JANTXV) of MIL-PRF-19500 and as follows. Electrical measurements (end points) shall be in accordance with the table I, group A, subgroup 2 herein.

4.4.2.1 Group B inspection, table VIa (JANS) of MIL-PRF-19500.

Subgroup	Method	Condition
B4	1037	1N483B, 1N5194, 1N5194UR, 1N5194US - V_R = 70 V (pk), 1N485B, 1N5195, 1N5195UR, 1N5195US - V_R = 180 V (pk), 1N486B, 1N5196, 1N5196UR, 1N5196US - V_R = 225 V (pk), I_o = 50 mA dc, f = 60 Hz, T_A = 25° C, t_{on} = t_{off} = 3 minutes minimum for 2,000 cycles.
B5	1027	See 4.3.1
B6	----	Not applicable.

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4.4.2.1 Group B inspection, table VIb (JAN, JANTX, and JANTXV) of MIL-PRF-19500.

Subgroup	Method	Condition
B3	1027	See 4.3.1.
B5	----	Not applicable.

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table VII of MIL-PRF-19500, and as follows. Electrical measurements (end points) shall be in accordance with the table I, group A, subgroup 2 herein.

Subgroup	Method	Condition
C2	2036	Lead fatigue: Test condition E.(Terminal strength not applicable to UR and US suffix devices.)
C6	1026	See 4.3.1

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables.

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TABLE I. Group A inspection.

Inspection 1/, 2/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 1						
Visual and mechanical inspection	2071					
Subgroup 2						
Thermal impedance	3101	See 4.4.1	$Z_{\Theta JX}$		70	°C/W
Forward voltage	4011	I_F = 100 mA (pk) (Pulsed); t_p = 8.5 ms max. duty cycle ≤ 2%.	V_{F1}	.8	1.0	V (pk)
Reverse current	4016		I_{R1}			
1N483B, 1N5194		V_R = 70 V (pk)			25	nA dc
1N485B, 1N5195		V_R = 180 V (pk)			25	nA dc
1N486B, 1N5196		V_R = 225 V (pk)			25	nA dc
Reverse current	4016		I_{R2}			
1N483B, 1N5194		V_R = 80 V (pk)			100	µA dc
1N485B, 1N5195		V_R = 200 V (pk)			100	µA dc
1N486B, 1N5196		V_R = 250 V (pk)			100	µA dc
Subgroup 3						
High temperature operation		T_A = 150° C				
Reverse current	4016		I_{R3}			
1N483B, 1N5194		V_R = 70 V (pk)			5	µA dc
1N485B, 1N5195		V_R = 180 V (pk)			5	µA dc
1N486B, 1N5196		V_R = 225 V (pk)			5	µA dc
Low temperature operation		T_A = -55° C				
Forward voltage	4011	I_F = 100 mA (pk) (Pulsed); t_p = 8.5 ms (max) duty cycle ≤ 2%.	V_{F2}		1.2	V (pk)
Subgroups 4 and 5						
Not applicable						
Subgroup 6						
Surge current	4066	I_o = 200 mA dc, 10 surges at one per minute; t_p =1/120 s, I_{FSM} = 2 A (pk).				
Electrical measurements		See table I, group A, subgroup 2				

1/ Electrical characteristics for UR and US suffix devices are identical to the corresponding non-UR or US suffix devices unless otherwise specified (Applies only to the 1N5194, 1N5195, and 1N5196).

2/ For sampling plan, see MIL-PRF-19500.

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of material is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Points' packaging activity within the Military Department or Defense Agency, or within the Military Departments' System Command. Packaging data retrieval is available from the managing Military Departments' or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

5.2 Marking. Unless otherwise specified (see 6.2), marking shall be in accordance with MIL-STD-129.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Notes. The notes specified in MIL-PRF-19500 are applicable to this specification.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of the specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1).

c. Lead finish (see 3.3.1).

CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - EC
Air Force - 17

Review activities:
Army - AR, MI
Navy - AS, CG, MC
Air Force - 19, 80, 85, 99
NASA - NA

Preparing activity:
DLA - CC

(Project 5961-1875)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	**1. DOCUMENT NUMBER** MIL-PRF-19500/118E	**2. DOCUMENT DATE (YYMMDD)** 97/04/23

3. DOCUMENT TITLE SEMICONDUCTOR DEVICE, DIODE, SILICON, TYPES 1N483B, 1N485B, 1N486B, 1N5194, 1N5194UR, 1N5194US, 1N5195, 1N5195UR, 1N5195US, 1N5196, 1N5196UR, AND 1N5196US JAN, JANTX, JANTXV, AND JANS

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME Alan Barone	b. TELEPHONE (Include Area Code) (1) Commercial 614-692-0510 (2) AUTOVON 850-0510
c. ADDRESS (Include Zip Code) Defense Supply Center Columbus ATTN: DSCC-VAT Columbus, OH 43216-5000	**IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:** Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

DD Form 1426, OCT 89 Previous editions are obsolete 198/290

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